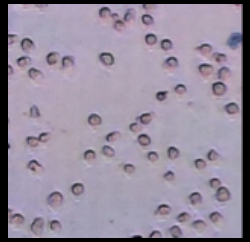
CELL IMAGE COUNTER



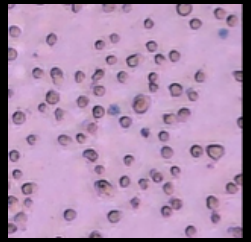
11.06.2026

Ruiping Huang, Dingshan Deng,
Pranjal Singh, Jungmin Seo

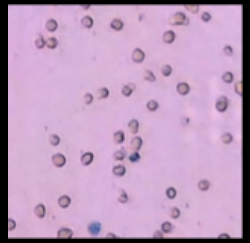
BACKGROUND



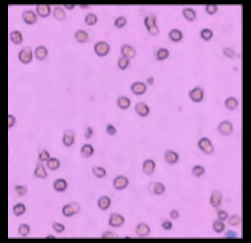
54 labeled cells



74 labeled cells



41 labeled cells



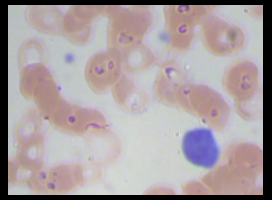
56 labeled cells

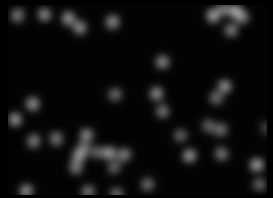
Image examples from: Roboflow cell counting

Manual cell counting is slow and repetitive: ~ 1 min per image

Counting challenges:

- Cell density
- Overlapping cells
- Image quality



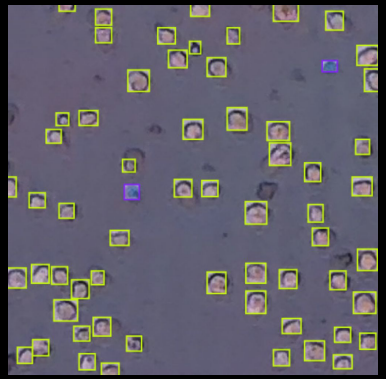


Potential users:

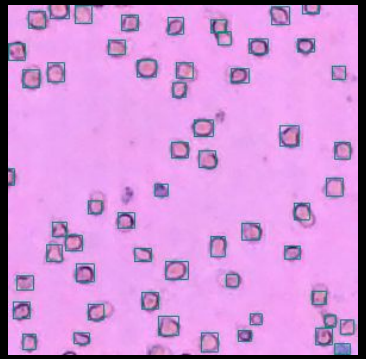
- Medical companies
- Pathology labs
- University research labs
- Private research labs

Our Goal:
Accurate and efficient automated cell counting

DATASET



Example 1



Example 2

Labels in Example 1:

- Dead: purple box
- Live: yellow box

500 original images

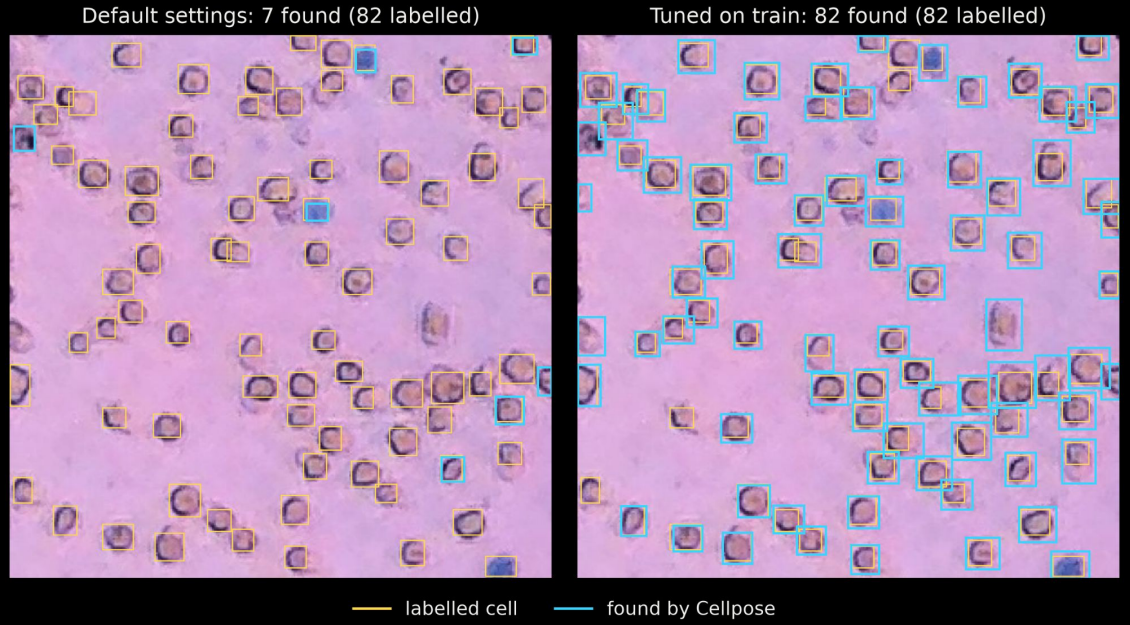
26,715 annotated cells

5 source groups

640 × 640 RGB microscopy images

Count Live and Dead cells together

Cell counting (2023). *cell counting Dataset*. Roboflow Universe. CC BY 4.0. https://universe.roboflow.com/cell-counting/cell-counting-zeqo5



We tune parameters
on train images only:
cell-probability threshold 0
→ −4 (keep faint pixels)

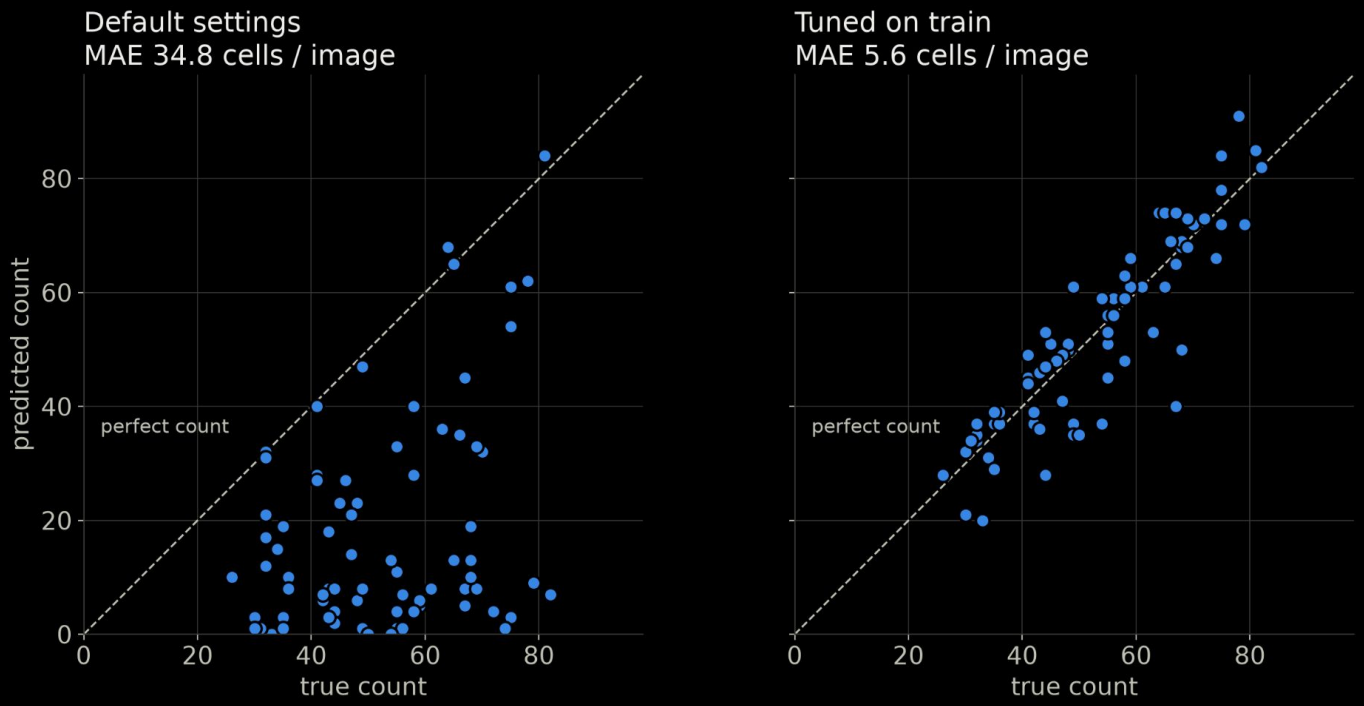
shape-check threshold 0.4
→ 1.0 (keep irregular outlines)

Fig: Same image, same network. Defaults find 7 of 82 cells; tuned settings find about all of them

BASELINE MODEL: CELLPOSE

cellpose
a generalist algorithm for cellular segmentation

A pre-trained cell segmenter, no training.
Pachitariu et al. (2025)
https://doi.org/10.1101/2025.04.28.651001



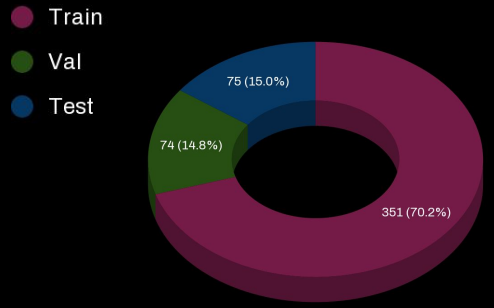
- ✅ ~30× faster than counting by hand (1.85 s/image)
- ⚠️ Misses pale, faint, and irregular cells

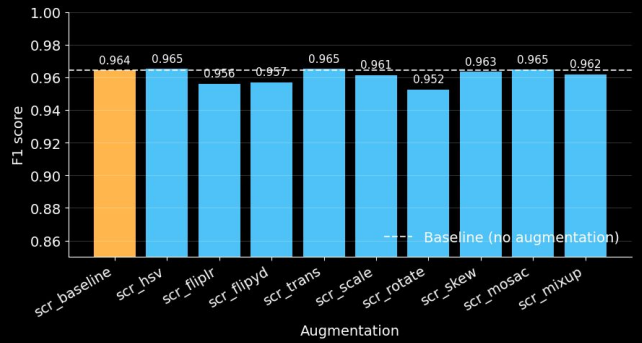
MODEL: ULTRALYTICS YOLO26



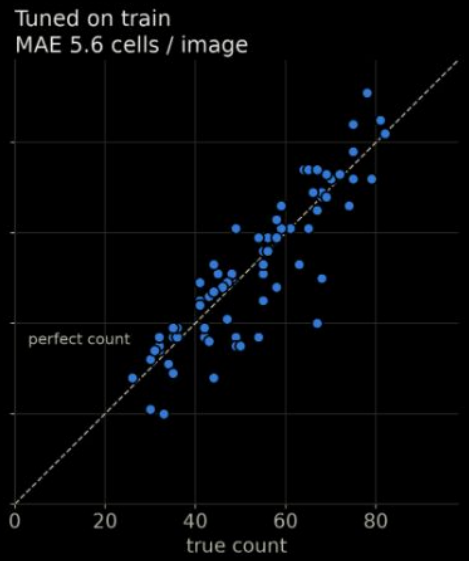
Jocher, G., Qiu, J., Liu, M., Lyu, S., Akyon, F. C., & Kalfaoglu, M. E. (2026). Ultralytics YOLO26: unified real-time end-to-end vision models. arXiv preprint arXiv:2606.03748

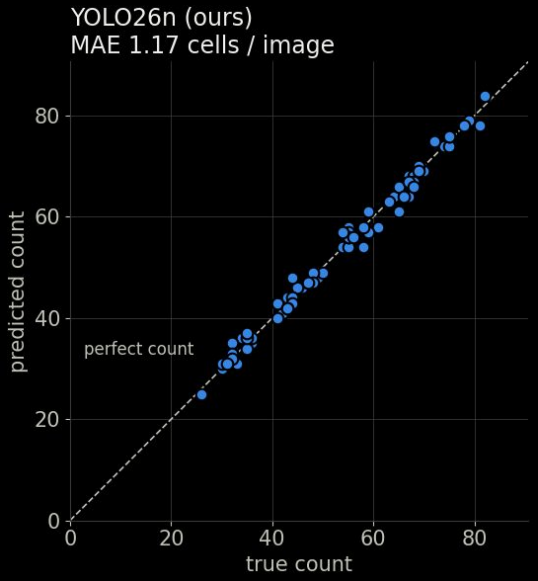
- Object detection is the primary task supported by YOLO26
- Standard Image Resolution: 640 (consistent with our data)

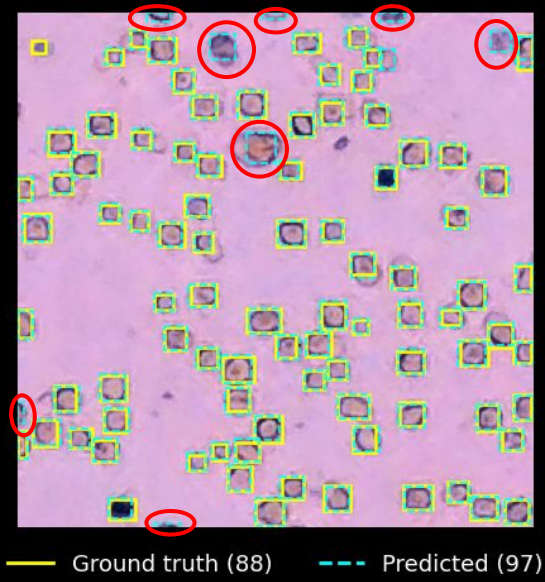










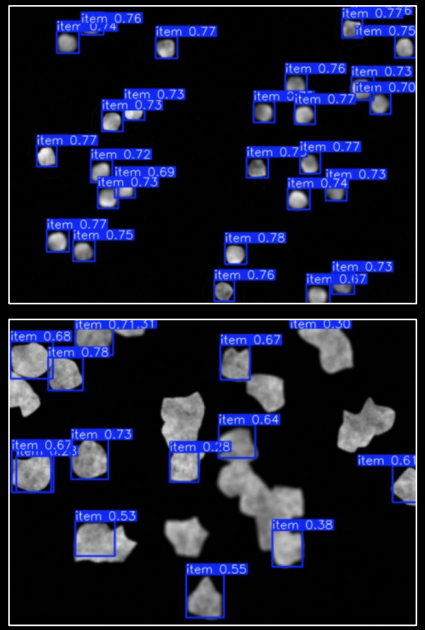


PERFORMANCE

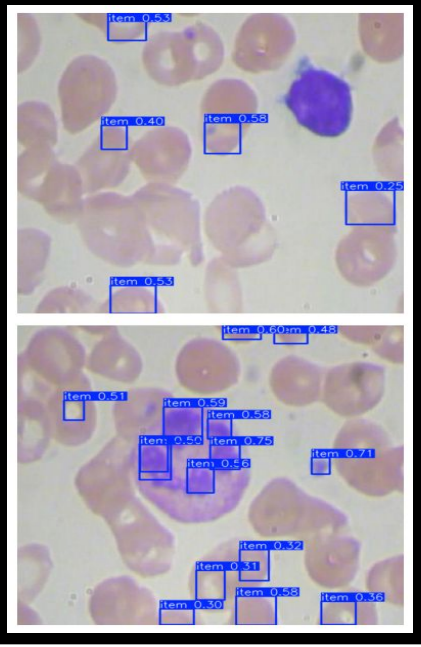
Metric	Cellpose-SAM (tuned)	YOLO26n (ours)	Improvement
MAE (cells)	5.6	**1.17**	**4.8× lower**
R^2	0.74	**0.989**	**+0.25**
Precision	0.91	**0.966**	**+5.6 pts**
Recall	0.90	**0.981**	**+8.1 pts**
Time / image	1.8 s	**28.6 ms**	**~63× faster**

Precision/recall: Cellpose on test set at IoU ≥ 0.3; YOLO on validation set at IoU ≥ 0.5.

Synthetic Dataset Performance



Blood Cell Performance



Tomato Detection Performance



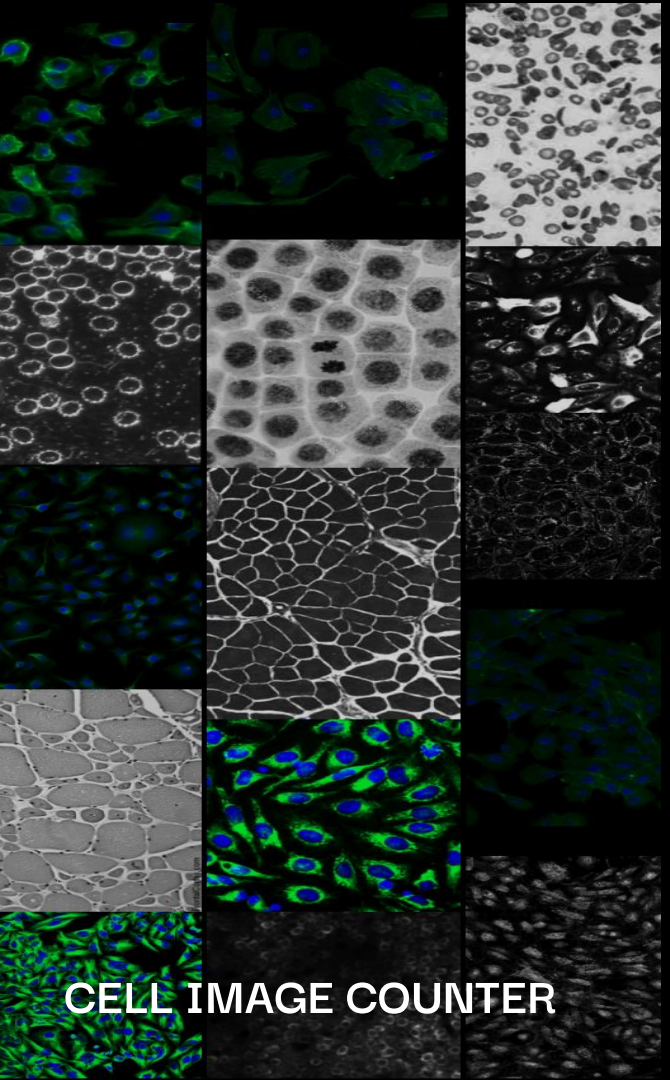
APPLICATIONS & IMPROVEMENTS

Egg Detection Performance



SUMMARY

- Our deep learning model provides a tool that allows users to count cells with the YOLO26 model, demonstrating a strong cell detection performance.
- Our model outperformed the general-purpose Cellpose-SAM baseline model, achieving a lower MAE (-80%), higher R^2 (+34%) and F1-score (+8%).
- There is a trade-off between a general-purpose model vs. task-specific model.
- Evaluating our fine-tuned YOLO26 model on different types of objects revealed that our model works best for crisp, circular, well-separated objects, appearing relatively small within the frame.
- Further investigation into model retraining with other types of cell datasets is needed.



THANK YOU

CELL IMAGE COUNTER